

Mathematical Analysis 2

Seminar 14

Double integrals (3)

$$\begin{cases} x = x(u,v) \\ y = y(u,v) \end{cases} ; (u,v) \in E \mapsto (x,y) \in D \Rightarrow \iint_D f(x,y) dx dy = \iint_E f(x(u,v), y(u,v)) |J(u,v)| du dv$$

$$dx dy = |J(u,v)| du dv, \quad J(u,v) = \det \begin{pmatrix} x'_u & x'_v \\ y'_u & y'_v \end{pmatrix}$$

(explicit)

$$\begin{cases} u = u(x,y) \\ v = v(x,y) \end{cases} ; du dv = |J(x,y)| dx dy ; J(x,y) = \det \begin{pmatrix} u'_x & u'_y \\ v'_x & v'_y \end{pmatrix}$$

(implicit)

$$J(x,y) = \frac{1}{J(u,v)}$$

Transforming (almost) everything into rectangles

Ex. 1 (parallelogram)

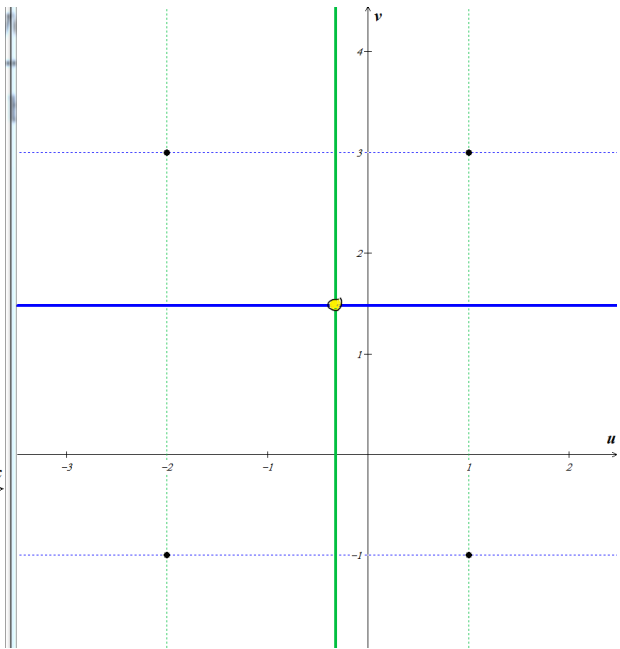
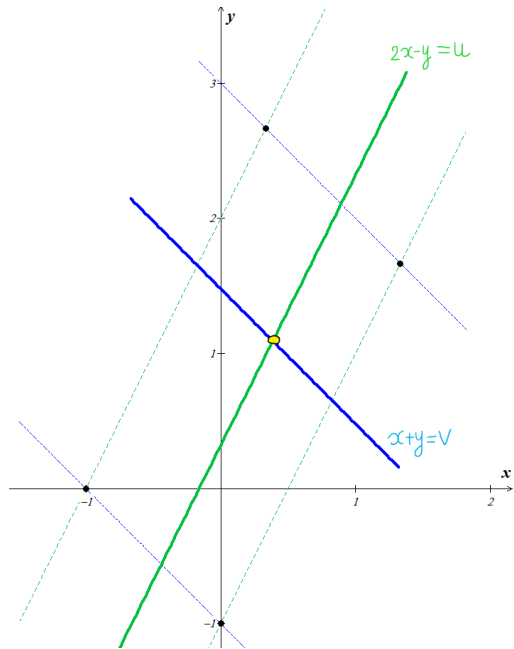
D bounded by $\begin{cases} 2x - y = -2, & x + y = -1 \\ \underbrace{2x - y = 1}_u, & \underbrace{x + y = 3}_v \end{cases}$

$$\iint_D (4x + y) dx dy =$$

$$D \mapsto E: \begin{cases} u = 2x - y \in [-2, 1] \\ v = x + y \in [-1, 3] \\ \text{(implicit)} \end{cases}$$

We solve the system: $\begin{cases} x = \frac{1}{3}u + \frac{1}{3}v \\ y = -\frac{1}{3}u + \frac{2}{3}v \\ \text{(explicit)} \end{cases}; J(u, v) = \begin{vmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{vmatrix} = \frac{2}{9} + \frac{1}{9} = \frac{1}{3}$

$$\begin{aligned} I &= \iint_E (u+2v) \cdot \frac{1}{3} du dv = \frac{1}{3} \int_{-2}^1 du \int_{-1}^3 (u+2v) dv = \frac{1}{3} \int_{-2}^1 (uv + v^2) \Big|_{v=-1}^{v=3} du = \frac{1}{3} \int_{-2}^1 (4u + 8) du = \\ &= \frac{1}{3} (2u^2 + 8u) \Big|_{-2}^1 = \dots \end{aligned}$$



Ex. 2 (trapezoid¹)

D bounded by

$$\begin{cases} x+2y=1 \\ x+2y=4 \end{cases} \quad \begin{cases} y=x \\ y=3x \end{cases} \quad \begin{aligned} u &\in [1, 4] \\ v &= \frac{y}{x} \in [1, 3] \quad (y=vx) \end{aligned}$$

$$\iint_D \frac{1}{x^2} dx dy =$$

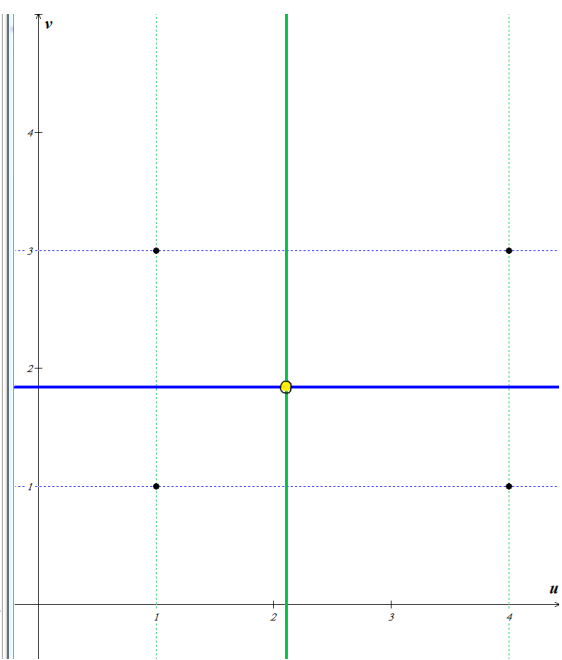
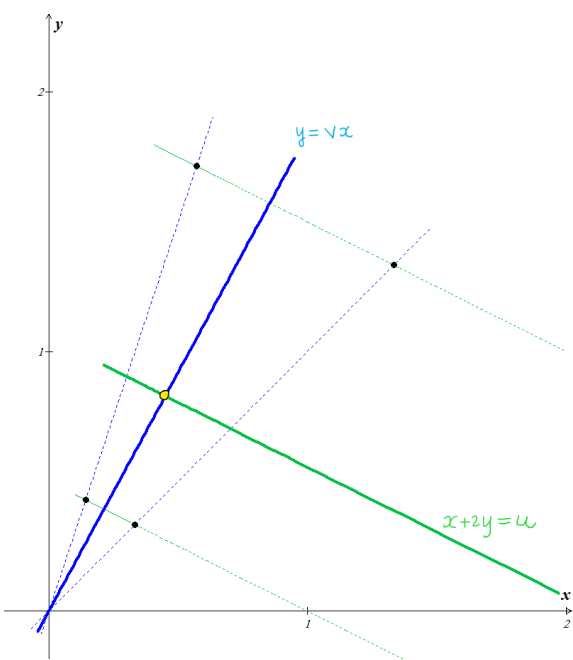
$$\begin{cases} u = x+2y \\ v = \frac{y}{x} \end{cases} \quad J(x,y) = \begin{vmatrix} u'_x & u'_y \\ v'_x & v'_y \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix} = \frac{1}{x} + \frac{2y}{x^2} = \frac{x+2y}{x^2} \quad ; \quad E = [\underbrace{1, 4}] \times [\underbrace{1, 3}]$$

(implicit)

$$\frac{1}{x^2} dx dy = \underbrace{\frac{1}{x^2} \frac{1}{|J(x,y)|}}_{\text{?}} \underbrace{|J(x,y)| dx dy}_{du dv} = \frac{1}{x^2} \cdot \frac{x^2}{|x+2y|} \cdot du dv = \frac{1}{|x+2y|} du dv = \frac{1}{|u|} du dv = \frac{1}{u} du dv$$

$$I = \iint_E \frac{1}{u} du dv = \left(\int_1^4 \frac{1}{u} du \right) \left(\int_1^3 dv \right) = \ln u \Big|_1^4 \cdot v \Big|_1^3 = 2 \cdot \ln 4 = 4 \ln 2.$$

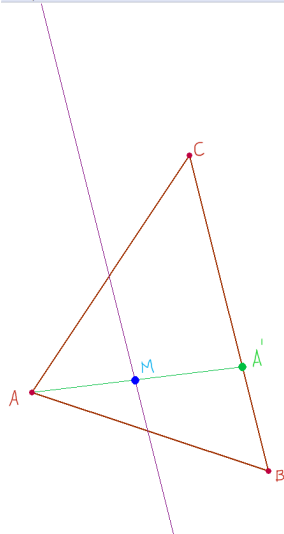
¹trapezoid (AmE) / trapezium (BrE)



Ex. 3 (triangle)

ma2_2020_s14_ex_3_triangle_1

File Equa View Btms One Two Anim Misc



$\iint_D (x-y) dx dy$, D is the interior of the triangle ABC ,
with $A(2, 1)$, $B(5, 0)$, $C(4, 4)$.

$$\vec{r}_{A'} = (1-t)\vec{r}_B + t\vec{r}_C, \quad t \in [0, 1] \quad (t = \frac{A'B}{BC}) \quad (s = \frac{AM}{AA'})$$

$$A' = (1-t)B + tC$$

$$M = (1-s)A + sA' = \dots \Rightarrow M = A + s(B-A) + st(C-B), \quad s \in [0, 1]$$

$$J(s, t) = s \cdot 2 \sigma(\Delta ABC) = s \cdot \Delta, \quad \Delta = \begin{vmatrix} 1 & 1 & 1 \\ x_A & x_B & x_C \\ y_A & y_B & y_C \end{vmatrix} = \begin{vmatrix} x_B - x_A & x_C - x_A \\ y_B - y_A & y_C - y_A \end{vmatrix}$$

$$B-A: (3, -1) \quad C-B: (-1, 4) \quad \begin{cases} x = x_A + s(x_B - x_A) + st(x_C - x_B) = 2 + 3s - st \\ y = y_A + s(y_B - y_A) + st(y_C - y_B) = 1 - s + 4st \end{cases}, s, t \in [0, 1]$$

$$J(s, t) = s \cdot \begin{vmatrix} 3 & -1 \\ -1 & 4 \end{vmatrix} = 11s; \quad x-y = 1 + 4s - 5st$$

$$D \rightarrow E: \begin{cases} s \in [0, 1] \\ t \in [0, 1] \end{cases}; \quad I = \iint_E (1 + 4s - 5st) \cdot 11s \, ds dt =$$

$$= 11 \int_0^1 ds \int_0^1 (s + 4s^2 - 5s^2 t) dt = 11 \int_0^1 \left(s + 4s^2 - 5s^2 \frac{1}{2} \right) ds = 11 \int_0^1 \left(s + \frac{1}{2} \cdot 3s^2 \right) ds$$

$$= 11 \left(\frac{1}{2} + \frac{1}{2} \right) = 11.$$

Remark:

G = the center of mass for $\triangle ABC$; $G = \frac{A+B+C}{3} = \left(\frac{11}{3}, \frac{5}{3}\right)$

$$\left. \begin{aligned} x_G &= \frac{\iint_D x \, dx \, dy}{\sigma(D)} \Rightarrow \iint_D x \, dx \, dy = \frac{11}{2} \cdot \frac{11}{3} \\ y_G &= \frac{\iint_D y \, dx \, dy}{\sigma(D)} \Rightarrow \iint_D y \, dx \, dy = \frac{11}{2} \cdot \frac{5}{3} \end{aligned} \right\} \ominus \Rightarrow I = \frac{11}{2} \cdot \frac{11-5}{3} = 11$$

$$\sigma(D) \left(= \iint_D dx \, dy \right) = \frac{1}{2} \begin{vmatrix} 1 & 1 & 1 \\ 2 & 5 & 4 \\ 1 & 0 & 4 \end{vmatrix} = \frac{11}{2}$$

Another method

D is simple w.r.t x and wrt y

$$x \in [2, 5]$$

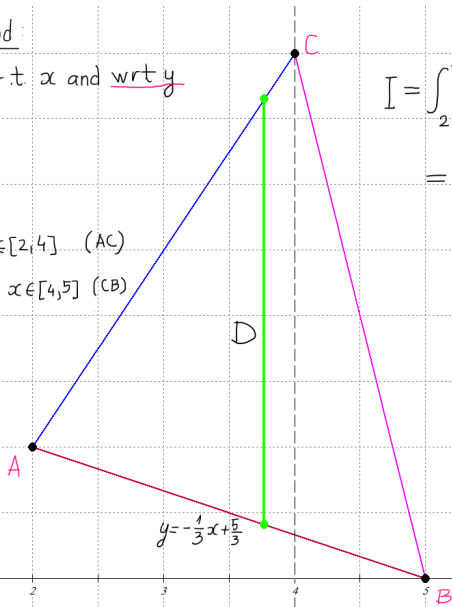
$$y \in [c(x), d(x)]$$

$$c(x) = -\frac{1}{3}x + \frac{5}{3}$$

$$d(x) = \begin{cases} \frac{3}{2}x - 2, & x \in [2, 4] \text{ (AC)} \\ -4x + 20, & x \in [4, 5] \text{ (CB)} \end{cases}$$

$$I = \int_2^5 dx \int_{c(x)}^{d(x)} (x-y) dy =$$

$$= \int_2^4 dx \int_{-\frac{1}{3}x + \frac{5}{3}}^{\frac{3}{2}x - 2} (x-y) dy + \int_4^5 dx \int_{-\frac{1}{3}x + \frac{5}{3}}^{-4x + 20} (x-y) dy$$



Ex. 4 D bounded by

$$\begin{cases} x^2 = y, \\ x^2 = 4y, \end{cases} \quad \begin{cases} y^2 = 2x \\ y^2 = 5x \end{cases}$$

$$\sigma(D) =$$

$$\frac{x^2}{y} = u \in [1, 4] \quad ; \quad \frac{y^2}{x} = v \in [2, 5] \mapsto E : \begin{cases} u = \frac{x^2}{y} \in [1, 4] \\ v = \frac{y^2}{x} \in [2, 5] \end{cases} \quad (\text{implicit})$$

$$J(x, y) = \begin{vmatrix} \frac{2x}{y} & -\frac{x^2}{y^2} \\ -\frac{y^2}{x^2} & \frac{2y}{x} \end{vmatrix} = 4 - 1 = 3 \Rightarrow J(u, v) = \frac{1}{J(x, y)} = \frac{1}{3}$$

(explicit)

$$\sigma(D) = \iint_D dx dy = \iint_E \frac{1}{3} du dv = \frac{1}{3} \sigma(E) = \frac{1}{3} \cdot (4-1) \cdot (5-2) = 3.$$

Ex. 5 D in the first quadrant, bounded by

$$\begin{cases} x^2 - y^2 = 1, & x^2 + y^2 = 3 \\ x^2 - y^2 = 2, & x^2 + y^2 = 5 \end{cases}$$

$$\iint_D xy \, dx \, dy =$$

(individual study)

[Hint : use the $\underbrace{u}$ implicit change of $\underbrace{v}$ variable]